

Celebrity Problem

Problem Definition

The Celebrity Problem is a problem in which there are n people at a party and we need to find the celebrity, if one exists.

Conditions for a Celebrity

- Everyone knows the celebrity.
- The celebrity knows nobody.

Representation Using Matrix

The information about who knows whom is represented using a matrix M .

$M[i][j] = 1$ means person i knows person j .

$M[i][j] = 0$ means person i does not know person j .

Example

Consider the following matrix:

0	1	1
0	0	1
0	0	0

Person 2 is the celebrity because person 0 knows person 2, person 1 knows person 2, and person 2 knows nobody.

Approach

1. Start with person 0 as the candidate.
2. Compare the candidate with every other person.
3. If the candidate knows another person, that candidate cannot be the celebrity, so select the other person as the new candidate.
4. After finding the possible candidate, verify that the candidate knows nobody and everyone else knows the candidate.
5. If the conditions are satisfied, the candidate is the celebrity; otherwise, no celebrity exists.

C Program

```
#include <stdio.h>

int findCelebrity(int M[][100], int n)
{
    int candidate = 0;

    for (int i = 1; i < n; i++)
    {
        if (M[candidate][i] == 1)
            candidate = i;
    }
}
```

```

    }

    for (int i = 0; i < n; i++)
    {
        if (i != candidate &&
            (M[candidate][i] == 1 || M[i][candidate] == 0))
            return -1;
    }

    return candidate;
}

int main()
{
    int n;
    int M[100][100];

    printf("Enter number of people: ");
    scanf("%d", &n);

    printf("Enter the matrix:\n");

    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
        {
            scanf("%d", &M[i][j]);
        }
    }

    int celebrity = findCelebrity(M, n);

    if (celebrity == -1)
        printf("No Celebrity Found\n");
    else
        printf("Celebrity is Person %d\n", celebrity);

    return 0;
}

```

Sample Input

```

Enter number of people: 3
Enter the matrix:
0 1 1
0 0 1
0 0 0

```

Sample Output

```

Celebrity is Person 2

```

Complexity Analysis

Time Complexity: $O(n)$. Candidate selection takes $O(n)$ and verification takes $O(n)$.

Space Complexity: $O(1)$, excluding the input matrix.